

Deburring

what is the meaning of "deburring"?

“Deburring” means **removing burrs**.

A **burr** is a small rough edge, raised bit, or unwanted leftover material that remains after cutting, drilling, machining, stamping, or grinding metal, plastic, or wood.

Example:

- If you cut a metal plate and the edge feels sharp and ragged, **deburring** is the process of smoothing or cleaning that edge.

People do deburring to:

- make parts safer to handle
- improve fit between parts
- improve appearance
- prevent damage or stress points

Common deburring methods:

- sanding
- filing
- grinding
- brushing
- chemical deburring
- thermal deburring

In simple words: **deburring = cleaning up rough leftover edges after manufacturing.**

If you want, I can also explain the difference between **burring, deburring, chamfering, and polishing.**

give me the catalogs for deburring equipments

I’m pulling together actual manufacturer catalogs rather than a generic equipment list, so you get something you can open and compare side by side.

Thought for 19s

Yep — here are solid **official catalog / brochure sources for deburring equipment** you can open and compare.

For **general deburring and sheet-metal machines**:

- **Timesavers** — product lineup for metal deburring and finishing, including the 12 Series, 22 Series, and manual deburring machines. Good starting point for broad catalog browsing.
- **NS Máquinas** — has a dedicated downloads page with PDF brochures for sheet deburring, edge rounding, tube deburring, and bench grinders.
- **LISSMAC** — strong for sheet-metal deburring, double-sided processing, slag removal, and wet/dry finishing systems.
- **ARKU** — useful both for machine brochures and for a practical guide to automated deburring methods; especially relevant for laser-cut, punched, plasma-cut, and heavy-plate parts.
- **WEBER** — brochures for compact deburring systems and higher-end plate processing machines. Good if you want both compact and industrial-class options.

If you want the quickest shortlist by use case:

- **Laser-cut / punched sheet metal**: Timesavers, ARKU, LISSMAC, NS Máquinas.
- **Heavy slag removal / flame-cut or plasma-cut plate**: ARKU, LISSMAC, WEBER.
- **Tube / profile / bench deburring**: NS Máquinas, LOESER.
- **Automotive wheel deburring**: LOESER.

My practical take:

- Start with **Timesavers** or **NS Máquinas** if you want an easy broad overview.
- Start with **ARKU** or **LISSMAC** if your focus is serious **sheet-metal production**.
- Start with **WEBER** if you want compact-to-advanced industrial brochure options.
- Start with **LOESER** if your parts are more **tube/profile/wheel-specific**.

I can also organize these into:

1. **manual vs automatic deburring equipment catalogs**, or
2. **a shortlist for your exact material** like stainless sheet, aluminum, tube, or flame-cut plate.

give me the manuals vs automatic deburring equipment catalogs and prices

I'm sorting this into two buckets and checking which vendors actually publish prices versus quote-only, because industrial machinery pages love hiding the ball.

Most makers publish brochures and specs, but not sticker prices. I'm grabbing a few public market listings too so the price section is honest instead of hand-wavy.

Thought for 32s

Here's a practical split between **manual** and **automatic** deburring equipment, plus what I could verify about **catalogs** and **prices**.

Manual deburring equipment catalogs

Timesavers 10 Series / Manual Grinder

This is a hand-guided machine aimed at low-volume or specialized parts. Timesavers describes it as a starter model and a cost-saving alternative for manual deburring, edge rounding, and finishing. Their brochure also positions the 10 Series as the manual option in the lineup.

NS Máquinas CL75 and related manual / multipurpose machines

NS Máquinas' downloads page is the main catalog hub, and the CL75 is one of the manual-style multipurpose machines for grinding and deburring components. It is more "operator-driven flexible finishing" than a full conveyORIZED automated sheet system.

Automatic deburring equipment catalogs

Timesavers 22 Series / 32 Series / 31 Series

Timesavers' automatic line covers conveyORIZED deburring and finishing for laser-cut parts, edge rounding, and oxide removal. The 22 Series is presented as a competitive entry automatic option, while the 32 and 31 series step up in capability and duty cycle.

LISSMAC SMD 123 RE

This is an automatic dry grinding/deburring machine for sheet metal. LISSMAC lists it as an economical entry-level system for deburring, oxide removal, and uniform edge rounding, with a 950 mm working width and 1–50 mm material thickness range.

ARKU automated deburring machines

ARKU's catalog content is focused on automated deburring for punched, laser, plasma, and flame-cut parts, including heavy-plate applications. Their guide explicitly contrasts automated deburring with manual hand grinding and emphasizes consistency and downstream productivity.

NS Máquinas DM series

NS Máquinas' sheet-deburring range includes conveyORIZED automatic machines such as the DM1100/DM1600 family for edge rounding, slag removal, and deburring of sheet parts.

What pricing looks like

For **new industrial deburring machines**, most manufacturers do **not publish list prices** on their official product pages and instead route buyers to "request a quote" or sales contact forms. That

was the case on the Timesavers, LISSMAC, and NS Máquinas pages I checked.

So the most honest way to talk about price is:

Category	Example models	Typical pricing visibility
Manual / operator-guided	Timesavers 10 Series, NS CL75	Mostly quote-only from maker; some used listings
Automatic entry-level sheet deburring	Timesavers 22 Series, LISSMAC SMD 123 RE, NS DM1100 C	Usually quote-only; used market sometimes visible
Automatic mid/high-end heavy-duty	Timesavers 31/32/41, NS DM1100 DC EVO / 2C EVO, ARKU Edgebreaker-class systems	Almost always quote-only

Public price references I could verify

I found **public used-market pricing**, but not reliable official new-machine MSRP.

- A Machinio listing page showed a **used Timesavers deburrer/grinder** at **\$40,000 USD**. That is a marketplace listing, not factory pricing.
- Another Machinio page showed an older **Timesavers 137-1HD** at **\$7,900 USD**, again used-market pricing.
- For **used LISSMAC metal cutting equipment** in the U.S., one listing source reported a current range around **\$19,500 USD** for the visible set it summarized, but that is an aggregate used-market snapshot, not a new deburring-machine list price.
- For several current **Timesavers 22 Series**, **LISSMAC SMD 123 RE**, and **NS Máquinas DM-series** listings, public marketplaces showed **“request price” / “contact seller”** rather than a posted amount.

My usable takeaway

If you want a rough buying ladder:

- **Manual / low-volume shop use:** expect the lowest capital cost, often found used in the **high four figures to tens of thousands USD** range, depending on age and condition.

- **Automatic entry sheet-metal deburring:** usually a serious capital machine and often **quote-only** when new; used units can show up in the **tens of thousands USD** range.
- **Heavy-duty automated slag removal / double-sided edge rounding:** typically quote-only and generally the most expensive tier. ARKU, higher-end LISSMAC, and larger NS/Timesavers systems live here.

If you want, I can turn this into a **clean vendor shortlist with columns for machine type, part size, material, automation level, and expected budget band**.

could you generate more detailed content of each manual technically?

I'm pulling the actual spec-level details for the manual-side machines so this is about working width, heads, materials, and use cases—not brochure fluff.

Thought for 18s

Yes — here's a more **technical breakdown of the manual-side deburring machines** I mentioned, focusing on **how they work, what heads/media they use, part handling, operating envelope, and where each machine fits**.

1) Timesavers 10 Series Manual Grinder

This is a **manual, operator-guided deburring and finishing station** for metal parts. Technically, it is built around a **two-head concept**: one side uses a **grinding disc / pad** for more aggressive stock removal or burr knockdown, and the other side uses a **brush-type head** for lighter deburring, edge softening, finishing, oxide removal, and some slag cleanup. The shaft rotates **180°**, so the operator can switch quickly between the two tool sides without changing machines.

From the spec sheet and product pages, the machine is intended for **manual deburring, edge rounding, and finishing** rather than conveyORIZED mass processing. The published working area is about **1300 × 800 mm**, with a **working height around 960 mm** on one spec sheet; one brochure lists **maximum part thickness 63.5 mm**, while another reseller page lists **100 mm**. I would treat **63.5 mm as the safer official brochure figure** unless the exact variant is confirmed by the seller. The machine also includes a **vacuum hold area** for smaller parts so they do not skate around during brushing or grinding.

How it works technically

The **disc side** is the more aggressive side. It is used for **pre-deburring**, removing standing burrs, and handling heavier edge cleanup. The **brush side** is then used to **break the sharp**

edge, create a more consistent radius, improve cosmetic finish, and handle lighter finishing tasks. Because different pads and brushes can be fitted, the machine is fairly flexible in abrasive behavior: coarse removal on one side, finer conditioning on the other.

Mechanically, the operator presents the part by hand onto the table/mat, controls the contact pressure manually, and moves the part relative to the abrasive head. That means **surface result quality depends partly on operator skill**, but it also makes the machine good for **fragile parts, odd geometries, prototypes, and low-volume work** where programming an automatic machine would be wasteful. Timesavers explicitly positions it for **specialized products** and **low-volume runs**.

Technical strengths

Its biggest technical advantage is **process flexibility**. A shop can use one station for:

- burr removal
- edge rounding
- cosmetic finishing
- laser oxide removal
- some heavy slag removal, depending on media selection.

Another advantage is that it is reportedly **3–4 times faster than hand deburring**, while still keeping the operator in direct control of pressure, part angle, dwell time, and local treatment. It also uses **long-life flap brushes**, which Timesavers highlights as a consumable-cost benefit.

Technical limitations

This is **not** a true high-throughput line machine. Because feed, contact, and dwell are manual, you should expect:

- lower repeatability than conveyorized automatic deburring
- more operator dependence
- slower throughput on large batches
- less uniform edge radius across hundreds of identical parts than a tuned automatic machine.

It is best viewed as a **manual process-improvement machine**, not a full automation cell. That's exactly why it makes sense for **single parts, small series, mixed jobs, and repair / rework environments**.

Best fit

Technically, the 10 Series fits shops doing:

- laser-cut sheet parts in small batches
 - mixed-material fab work
 - prototype and job-shop work
 - fragile or awkward parts that are risky in automatic feed systems
 - secondary finishing near welding or fabrication benches.
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2) NS Máquinas CL75

The CL75 is a **manual-operation multipurpose abrasive belt machine**. Unlike the Timesavers 10 Series, which is built around a disc-plus-brush arrangement, the CL75 is centered on a **75 × 2000 mm abrasive belt** driven by a **3 kW motor**. NS describes it as suitable for **finishing, polishing, grinding, and deburring** of different metals and components, with **contact roller grinding** and **top-belt flat-section deburring** as standard functions.

How it works technically

This machine uses a **belt grinding architecture**, which changes the process behavior a lot compared with a disc/brush system. The **contact roller** gives a more concentrated, controlled abrasive contact area, which is useful for:

- weld seam grinding
- local stock removal
- tube/profile work
- edge cleanup on formed parts.

The **top abrasive belt section** allows more planar deburring and surface work on flatter components. In practice, that means the CL75 is less of a dedicated “sheet deburring table” and more of a **general metal finishing base machine** that can deburr as one of several operations. NS also notes that **centerless and notching modules** can be connected, which expands it beyond simple flat-part deburring into a modular finishing platform.

Published specs

The page lists:

- **belt size:** 75 × 2000 mm
- **belt motor:** 3 kW
- **machine dimensions:** 1150 × 650 × 1100 mm
- **weight:** 80 kg.

These specs tell you something important: this is a **compact manual workstation**, not a floor-scale automated deburring cell. The narrow belt width especially suggests it is better for **targeted work on components, welds, bars, tubes, and smaller flat sections** than for mass deburring wide laser-cut sheets.

Technical strengths

The CL75's strength is **versatility per square foot**. One manual machine can do:

- deburring
- weld polishing
- grinding
- basic finishing
- some tube/profile work
- future module expansion.

Because it is a belt machine, it can also be a better choice than a brush machine when the user needs **more directional abrasion** or more focused weld cleanup. If your shop has mixed fabricated parts rather than just stacks of laser-cut blanks, this style can be very efficient.

Technical limitations

The tradeoff is that the CL75 is **less specialized for uniform edge rounding across broad flat sheet batches** than a dedicated automatic deburring machine, and arguably less optimized for “quick brush both sides of a small flat blank” than a dedicated manual deburring table like the Timesavers 10 Series. Its process is more manual, more localized, and more dependent on operator handling and technique.

Best fit

Technically, the CL75 makes the most sense when the shop does:

- mixed fabrication
 - weld cleanup
 - small-part deburring
 - tube and profile finishing
 - job-shop work where one machine must do several abrasive tasks.
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3) NS Máquinas CLF75

The CLF75 is another **manual belt-based machine**, but it is aimed at a different problem: **large, heavy plates that are difficult to bring to a fixed machine**. NS describes it as a **flat surfaces belt grinder** and specifically a **movable machine for grinding heavy plates**. In other words, the machine is brought to the workpiece, not the other way around.

How it works technically

This is basically a **mobile belt grinding platform** for big flat metal surfaces. It uses:

- a **75 × 2000 mm abrasive belt**
- **30 m/s belt speed**
- a **Ø200 × 75 mm contact roller**
- **4 kW motor / 4 kW total power.**

That setup is technically suited for:

- surface cleaning on large plate
- calamine/scale cleanup
- weld seam grinding on wide panels
- belt finishing on oversized sheets
- deburring work where the part is too large or too heavy for standard handling.

Because it is wheel-mounted and intended to move along wide sheets, the operator can run the machine across the work surface while using the contact roller for controlled abrasive contact. This is very different from bench-style deburring: it is closer to a **mobile surface conditioning grinder** for fabrication floors.

Published specs

The brochure lists:

- **belt speed:** 30 m/s
- **belt dimensions:** 75 × 2000 mm
- **contact roller:** Ø200 × 75 mm
- **motor:** 4 kW
- **total power:** 4 kW
- **machine dimensions:** 1420 × 720 × 1010 mm
- **weight:** 105 kg
- **current draw:** 9 A.

Technical strengths

Its core advantage is handling **oversized floor-level workpieces**. NS explicitly calls out:

- easy movement along wide metal sheets
- wheel level adjustment
- weld seam grinding with the contact roller
- belt grinding when the sheet is too big or too heavy to handle conventionally.

So if the job is **big plate, big welded panels, shipyard/fabrication style work, or heavy steel pieces**, the CLF75 is technically a much better match than a table-style manual deburrer.

Technical limitations

This is **not** the machine for precision small-part edge rounding or efficient small-batch deburring of many cut blanks. It is optimized for **surface grinding and seam finishing on large work**, so it is overkill and awkward for tiny parts. It also remains fully manual, so finish consistency still depends on operator control and technique.

Best fit

Technically best for:

- heavy steel plates
- wide panels
- oversized welded fabrications
- flat-surface cleanup on the shop floor
- situations where the workpiece cannot reasonably be fed into a fixed machine.

Practical comparison

If I strip the marketing away and say it plainly:

Timesavers 10 Series

- Best as a **manual flat-part deburring / edge-softening / finishing station**
- Better for **small batches, mixed flat parts, fragile pieces**
- More “deburring-focused” than the NS belt machines.

NS CL75

- Best as a **compact general-purpose belt-finishing machine**
- Better for **mixed metalwork, welds, profiles, and component-level grinding/deburring**

- More “multipurpose abrasive workstation” than dedicated deburring station.

NS CLF75

- Best as a **mobile belt grinder for very large plates and panels**
- Better for **heavy fabrication and oversized workpieces**
- Least suited to small precision parts, best for **floor-level large-part processing**.

My blunt recommendation

If your target is specifically **deburring laser-cut or punched parts by hand**, start with the **Timesavers 10 Series**.

If your shop needs **one manual abrasive machine for many tasks**, the **CL75** is more flexible.

If your parts are **big plates or welded panels**, the **CLF75** is the right style.

I can also make you a **technical comparison table** with columns like:

machine architecture, abrasive type, contact method, workpiece size, burr-removal aggressiveness, edge-rounding quality, weld-seam capability, and best industry use.

yes, please

I’m turning it into a shop-floor comparison table so you can see which machine is actually for flat parts, mixed fab, or heavy plate.

Thought for 12s

Absolutely — here’s a **technical comparison table** for the three **manual-style deburring / finishing machines**.

Technical comparison table

Mach ne	Machi ne archit ectur e	Abras ive / tool type	Conta ct meth od	Best workp iece type	Debur ring aggre ssive ness	Edge- round ing qualit y	Weld seam capab ility	Flat- part suitab ility	Large /heav y plate suitab ility	Main techni cal stren gths	Main techni cal limitat ions
Time saver s 10 Serie s	Manu al bench -top debur	Disc / cup style heads ;	Opera tor- guide d conta	Small to mediu m flat metal	Medi um to high depend ing	Good for manu al edge	Limited to mode rate; possi	Excellent for manu al flat-	Poor for oversi zed	Stron g fit for manu al	Manu al proce ss, so repeat

Manual Grinder	ring station with rotating head assembly	marked for deburring, edge rounding, finishing, laser oxide removal, and heavy slag removal	cut on worktable; rotating grinding attachment 180°; vacuum area for smaller parts	parts, fragile parts, prototypes, low-volume jobs	on media and side used; intended for rough burr removal through finishing	softening/rounding on flat parts	ble for localized cleanup, but not mainly a weld-seam machine	part work	heavy plate	deburring of laser-cut/punch-d parts; vacuum area helps with small parts; compact bench format; good for mixed low-volume work	ability depends on operator; much lower throughput than conveyORIZED automatic machines; published maximum thickness varies by source between 63.5 mm and 100 mm, so exact variant
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											should be confirmed before buying
NS Máquinas CL75	Manual multipurpose abrasive belt machine	75 × 2000 mm abrasive belt	Contact roller grinding or top-belt flat-section deburring	Mixed fabricated parts, bars, profiles, smaller flat sections, weld areas	Medium; stronger for focused belt grinding than broad-surface edge conditioning	Moderate; can deburr edges, but it is more a multipurpose belt machine than a dedicated edge-rounding station	Good; explicitly suited for weld seam polishing	Good for smaller flat sections, but less specialized than a dedicated flat-part deburring table	Poor to fair for very large heavy plate	Very versatile for one-machine shops; supports grinding, deburring, weld polishing, and optional centerless/notching modules	Narrow 75 mm belt means it is not optimized for fast, uniform deburring of broad sheet batches; finish uniformity depends heavily on operator

											handli ng
NS Máquina CLF7 5	Mobil e flat- surfacc belt grinder on wheels	75 × 2000 mm abrasive belt, Ø200 × 75 mm contact roller	Movable machine brought to the workpiece; floor-level belt grinding	Large, heavy plates, , wide panels, oversized welded fabrications	Medium to high for surface clean up and seam work on big parts	Low to moderate for precision edge rounding; that is not its main job	Strong; explicitly intended for weld seam grinding/ polishing on large workpieces	Fair for big flat surfaces, but not ideal for small loose parts	Excellent	Best match when the part is too large or heavy to move into a fixed machine; built for wide sheet/ panel work and floor processing	Awkward for small parts and precision edge conditioning ; still manual, so consistently depends on operator technique; more of a heavy- fabrication surface machine than a dedic

												ated small- part debur rer
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Key specs at a glance

Machine	Power	Working area / belt size	Weight	Notable technical details
Timesavers 10 Series	About 2.4–2.5 kW total , depending on source/version	1300 × 800 mm table; vacuum area about 300–310 × 250–260 mm	About 255–270 kg	2 heads listed on one product page; 180° rotating attachment; manual flat-part station
NS CL75	3 kW belt motor	75 × 2000 mm belt	Product page emphasizes compact format	Contact roller grinding plus top-belt flat deburring; optional centerless/notching modules
NS CLF75	4 kW main / total power	75 × 2000 mm belt; 30 m/s belt speed	About 97–105 kg depending on source	Mobile machine for heavy plates; contact roller Ø200 × 75 mm ; wheel-mounted floor use

Which one to choose

If your real job is **manual deburring of laser-cut flat parts**, the **Timesavers 10 Series** is the cleanest fit. It is the most clearly focused on manual deburring, edge rounding, and flat-part finishing.

If you need **one manual abrasive machine for many shop tasks** like deburring, weld polishing, grinding, and occasional flat-section work, the **NS CL75** is the more flexible general-purpose option.

If your parts are **big, heavy plates or welded panels**, the **NS CLF75** is the right category because it is designed to move across the workpiece instead of forcing the workpiece into a fixed station.

My blunt recommendation by shop type

- **Sheet-metal fab shop, small batches: Timesavers 10 Series**
- **General fabrication / mixed metalwork shop: NS CL75**
- **Heavy fabrication / plate work / big welded panels: NS CLF75**

I can also make the next version as a **buyer's matrix with materials like stainless, aluminum, carbon steel, laser-cut parts, tube, and heavy plate**.

yes, please

I'm mapping these machines against actual shop materials and part types now, because that's what really decides whether a machine helps or becomes an expensive annoyance.

Thought for 8s

Here's a **buyer's matrix** for the three manual-side machines, organized by **material and part type** so you can match the machine to actual shop work.

Buyer's matrix

Shop material / part type	Timesavers 10 Series	NS Máquinas CL75	NS Máquinas CLF75	Best pick
Stainless steel flat parts	Very good for manual deburring, edge rounding, finishing, and laser oxide removal on flat metal parts. Good when batches are	Good for deburring and finishing stainless components, especially when parts also need weld polishing or localized grinding. Less specialized for	Usually not the right first choice unless the stainless parts are very large sheets or welded panels on the floor.	Timesavers 10 Series

	small or parts are fragile.	uniform flat-part edge rounding.		
Aluminum flat parts	Good fit for small-batch manual deburring and edge softening, especially where gentle operator control matters.	Good if the work is mixed and you also need general finishing or weld cleanup, but not as purpose-built for flat blank deburring.	Usually unnecessary unless the aluminum workpieces are very large floor-handled sheets.	Timesavers 10 Series
Carbon steel laser-cut blanks	Strong match. Timesavers explicitly positions it for deburring, edge rounding, laser oxide removal, and even heavy slag removal with different consumables.	Can do deburring, but it behaves more like a multipurpose belt grinder than a dedicated flat blank deburring station.	Better only when the steel parts are too large to move easily.	Timesavers 10 Series
Mixed fabricated parts	Good when many parts are still basically flat and need deburring/finishing by hand.	Best all-rounder here. The CL75 is explicitly a multipurpose abrasive belt machine for finishing, weld seam polishing, grinding, and deburring different metals and components.	Not ideal unless the “mixed work” includes many large heavy panels.	NS CL75
Tube, bar, profile, formed	Usable only in limited cases; it is	Best of the three. Its belt/contact-	Not appropriate for typical	NS CL75

parts	mainly a flat-part manual deburring station.	roller format and optional centerless/notching modules make it much more suitable for profiles and varied component geometry.	tube/profile work.	
Weld seam cleanup	Can help for localized cleanup, but that is not its main specialty.	Strong fit. NS specifically calls out weld seam polishing on the CL75.	Also strong, especially on large panels and heavy plates. NS explicitly positions CLF75 for weld seam grinding on wide panels.	CL75 for normal fab parts; CLF75 for very large panels
Heavy plate / oversized sheets	Poor fit. It is a table-style manual station, not a floor machine for massive workpieces.	Limited fit. It is compact and flexible, but not built around moving across large plates.	Best fit. The CLF75 is explicitly a movable floor machine for large, heavy metal sheets and plates, with wheel-mounted mobility and weld seam grinding capability.	NS CLF75
Prototype / low-volume / fragile parts	Best fit. Timesavers explicitly says manual deburring is useful for	Good if the prototypes also need grinding/polishing /weld work, but	Usually not the right category.	Timesavers 10 Series

	specialized products, low-volume runs, and parts too fragile for automatic equipment.	less targeted at flat-part deburring.		
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Plain-English recommendations

For a **sheet-metal fab shop** doing stainless, aluminum, or carbon-steel **flat cut parts**, the **Timesavers 10 Series** is the cleanest manual choice because it is built around manual deburring, edge rounding, finishing, laser oxide removal, and low-volume/fragile-part work.

For a **general fabrication shop** that needs one machine to handle **deburring, weld polishing, grinding, and mixed metal components**, the **NS CL75** is the best all-rounder. Its contact-roller and top-belt setup make it the most versatile of the three for non-flat-only work.

For **heavy fabrication**, big panels, or parts that are just annoying to move, the **NS CLF75** is the right category because it is designed to be moved across the workpiece on the floor.

Super-short pick guide

- **Mostly flat laser-cut parts: Timesavers 10 Series**
- **Mixed shop work and weld cleanup: NS CL75**
- **Heavy plates and wide welded panels: NS CLF75**

I can do one more step and make this into a **scored decision matrix** with columns like budget, floor space, operator skill, throughput, finish consistency, and maintenance.

give me the price for each model

I'm checking current public pricing for each exact model and separating official pricing from used-market listings so the numbers don't get mashed together.

Thought for 16s

Here's the honest version: **I could not verify public official new prices** for these three models. The manufacturer pages I checked for **Timesavers 10 Series**, **NS CL75**, and **NS CLF75** are all **quote-only / ask-for-a-quote** rather than published MSRP.

So the best current public pricing I could find is:

Model	New official price	Public price I could verify	Notes
Timesavers 10 Series Manual Grinder	Not publicly listed	\$6,459.17 USD for a used 2021 Timesavers 10-Series-1300-MG listing on MachineTools; other listings show call / request price only.	Treat this as used-market reference , not factory price.
NS Máquinas CL75	Not publicly listed	No public fixed price found ; NS pages show Ask for a Quote .	I could verify specs and quote-only availability, but not a posted selling price.
NS Máquinas CLF75	Not publicly listed	\$3,334.00 USD for a used 2014 CLF75 listing on ThisMachinery; retailer/manufacturer-style pages I found do not show a posted price.	Again, this is used-market reference , not new MSRP.

My takeaway:

- **Timesavers 10 Series:** public used reference around **\$6.5k**.
- **NS CL75:** **quote only** from sources I could verify.
- **NS CLF75:** public used reference around **\$3.3k** for an older 2014 machine.

One caution: used industrial machine listings can vary a lot by **year, tooling included, condition, voltage, dust extraction setup, and region**, so these numbers are best treated as **rough market signals**, not purchase-ready quotes.

I can also make you a **budget estimate table** with realistic bands like **used / refurbished / new-quote expected range** for each model.

which one is made in china?

I'm checking the manufacturers' company info and location details so I don't guess based on reseller pages or domain names.

Thought for 16s

From the models we discussed, **none of them appear to be made in China** based on the manufacturer information I could verify.

- **Timesavers 10 Series:** Timesavers is headquartered in **Maple Grove, Minnesota, USA**. Their U.S. site explicitly says some machines are “engineered, designed and built in the USA,” and their company pages identify them as a U.S.-based manufacturer. They also have a Netherlands/China presence for service and sales, but that is not the same as saying the 10 Series is Chinese-made.
- **NS Máquinas CL75:** NS Máquinas is based in **Gondomar, Portugal**. Their site says the company is located in Portugal, and one of their about pages says **all machines are designed in-house in Portugal** and shows them as built there.
- **NS Máquinas CLF75:** same manufacturer, so also **Portugal-based**, with the same Portugal company/manufacturing context.

So the practical answer is:

- **Timesavers 10 Series:** U.S./international brand, not verified as China-made.
- **NS CL75:** Portugal.
- **NS CLF75:** Portugal.

If you want, I can next give you **manual deburring machine brands that are actually made in China** and compare them against these three.

yes, please

I'm checking for China-based brands with manual-style deburring or abrasive finishing machines, not just random Alibaba listings with vague OEM labels.

Thought for 12s

Yep — if you want **manual-style deburring machines that appear to be made in China**, here are the clearest ones I could verify from current public listings and company pages.

The three models we discussed earlier are **not** the China-made ones:

- **Timesavers 10 Series** is from Timesavers, a U.S.-based manufacturer.
- **NS Máquinas CL75 / CLF75** are from NS Máquinas in Portugal. (From earlier checked manufacturer info.)

The **China-made options I could verify** are these:

1. **Jinan D. Ventus manual grinding / deburring machine for flat sheets**

Made-in-China listings identify **Jinan D. Ventus Mechanical Equipment Co., Ltd.** as the manufacturer, and the products are explicitly presented as **manual grinding/deburring machines** for sheet metal, including aluminum and stainless applications. One listing is a **“Precision Manual Grinding Deburring Machine for Flat Sheets with Polishing Function.”**

2. **Jinan Knoppo manual grinding and deburring machine**

A Made-in-China listing shows **Jinan Knoppo Automation Equipment Co., Ltd.** offering an **“Easy-to-Operate Manual Grinding and Deburring Machine for Small Batches of Workpieces”** with a public listed price of **US\$3,999**. That is a China-based supplier and a manual-style deburring machine listing.

3. **RSM Machinery manual deburring machines**

Made-in-China shows **RSM Machinery** with a dedicated **“Manual Deburring Machine”** catalog page and example pricing around **US\$5,000–5,500 FOB** for stainless/sheet-metal grinding-deburring equipment. This is one of the clearest China-sourced manual-deburring catalog groupings I found.

If you also count **Chinese manufacturers of deburring equipment more broadly**, not just the exact manual bench-style format, there are several more:

4. **Hangzhou Xiangsheng / Jonsen Sander**

A Made-in-China company page says **Hangzhou Xiangsheng Abrasive Machine Manufacturing Co., Ltd. (Jonsen Sander)** was founded in **Hangzhou, Zhejiang, China**, and it sells abrasive belt grinding and deburring machines. The specific listing I found was more gantry/automatic-industrial than manual, but the maker is clearly China-based.

5. **Qingdao Xinkaide Machinery Manufacture Co., Ltd.**

Their company page says they are in **Qingdao, Shandong, China** and they manufacture **wide belt abrasive and brushing machines**, including **deburring machines** and belt grinders. Again,

more industrial-line than small manual bench model, but definitely a China manufacturer in this category.

6. **NEWSTEC / Shandong-based supplier**

An Alibaba product page identifies **NEWSTEC** as a **Chinese manufacturer based in Shandong** offering metal deburring and sanding equipment. This is useful as a China-source lead, though the page is more sales-listing style than a polished official catalog.

If your actual question is “**which of the manual-style machines similar to Timesavers 10 are made in China?**”, the most relevant answers are:

- **Jinan D. Ventus**
- **Jinan Knoppo**
- **RSM Machinery**

If you want the most practical next step, I can turn these into a **China-made shortlist table** with columns for **manufacturer, machine type, manual vs automatic, price, width, material suitability, and supplier link quality**.

give me the manual for these models and price

I’m looking for actual manuals or brochure PDFs for those China-made manual machines, plus any posted prices that are public right now.

Thought for 13s

I found **product manual/spec pages** for the China-made manual deburring models we discussed, plus the **public prices I could verify**. I did **not** find downloadable operator manuals for all of them, so below I’m giving you the closest reliable thing available publicly: the manufacturer’s spec/manual page or listing page.

China-made manual deburring models: manual/spec page + price

Model	Public manual / spec page	Public price I could verify	Notes
Jinan Knoppo KD-BB1308D Manual Deburring Machine	Knoppo product/spec page with dimensions and operating specs: workbench 1300 × 800 mm , vacuum area	I could verify a public Knoppo manual-type deburring listing at US\$3,999 for an “Easy-to-Operate	Best verified source for specs is Knoppo’s product page. The price is public, but

	300 × 250 mm, grinding plate 150 mm, head rotation 180°, max workpiece thickness 60 mm, machine weight 270 kg.	Manual Grinding and Deburring Machine for Small Batches of Workpieces.” The exact product title in that listing is a generic Knoppo manual deburring machine rather than the KD-BB1308D model code, so treat it as a brand/model-family price signal, not a confirmed KD-BB1308D quote.	model matching is not perfect.
RSM RMD-150 Manual Deburring Machine	RSM product page with structure/features for the RMD-150, including adjustable grinding head speed, quick-change head system, aluminum alloy arm, and flat-part / small curved-surface use.	US\$3,500 / piece FOB for RSM’s “Upgraded 3rd Generation Manual Deburring Machine, 1–150 mm Vacuum Adsorption Table Edge Rounding Polishing Machine for Metal Sheet.” I could also verify Alibaba pricing for an RSM robot-arm style manual sheet metal polishing/deburring machine at US\$3,600–3,700 per piece.	The public prices appear to be for RSM manual-deburring machines in this family, but not every listing explicitly says “RMD-150.”
RSM RMD-165 Manual Deburring Machine	RSM product page for RMD-165, described as a semi-automatic	I did not find a public listing that clearly names RMD-165 with	This one looks to be quote-only from what I found.

	middle-ground option for laser, plasma, and punch parts.	a fixed price. The only public RSM prices I could verify were the broader manual-deburring-family prices above.	
Jinan D. Ventus manual grinding / deburring machine	Made-in-China product/spec page for “Aluminum Stainless Steel Chamfering Equipment Manual Grinding Deburring Machine” from Jinan D. Ventus, with double abrasive belt frame details for stainless and carbon steel use.	I could verify the D. Ventus machine page, but I could not verify a public fixed price on the source I checked. It appears to be inquiry / quote based .	Good public spec page, but no reliable posted price surfaced in the sources I checked.

Direct links to the public “manual/spec” pages

- **Knoppo KD-BB1308D Manual Deburring Machine:** official product/spec page.
- **RSM RMD-150 Manual Deburring Machine:** official product/spec page.
- **RSM RMD-165 Manual Deburring Machine:** official product/spec page.
- **Jinan D. Ventus manual grinding/deburring machine:** public product/spec page.

My plain take

The most solid combo of **clear specs + public price** is **RSM**. I could verify product pages for the RMD series and separate public price listings in the **US\$3,500 to US\$3,700** range for their manual deburring machines.

For **Knoppo**, the specs are clearer than the pricing. I could verify the **KD-BB1308D** specs directly, but the **US\$3,999** price I found appears tied to a generic Knoppo manual deburring listing rather than the exact KD-BB1308D page.

For **D. Ventus**, I could verify a public model/spec page, but not a trustworthy fixed public price.

I can also turn this into a **clean procurement table** with columns for **exact specs, quoted/public price, source type, and confidence level**.

yes, please

I’m tightening this into a procurement-style table with exact model names, source type, and how confident each price match really is.

Thought for 10s

Absolutely — here’s a **clean procurement table** for the China-made manual/manual-style deburring models we discussed, with **spec source**, **public price**, **source type**, and **confidence** about the price-to-model match.

Procurement table

Manufacturer	Model	Public spec/manual source	Public price I could verify	Source type	Price-match confidence	Notes
Jinan Knoppo	KD-BB1308D	Knoppo product pages identify the KD-BB1308D Manual Deburring Machine and describe it as a manual deburring machine with vacuum table for deslagging, deburring, edge rounding, oxide	US\$3,999 for a Knoppo “Easy-to-Operate Manual Grinding and Deburring Machine for Small Batches of Workpieces.”	Marketplace listing	Medium	Good public price signal for the brand and machine family, but the listing snippet does not explicitly show the KD-BB1308D model code.

		removal, and polishing.				
RSM Machinery	RMD-150	RSM's product page identifies the RMD-150 as an entry-level manual deburring machine for small flat sheet-metal parts, with adjustable speed and grinding heads. Public spec listings show 5.8 kW power, 2 kW grinding head power, 40 kPa adsorption pressure, 1100 mm / 360° cantilever range , and Ø150 mm	US\$3,500 / piece FOB on Made-in-China for an RSM upgraded manual deburring machine, and Alibaba also shows an RMD-150 listing page.	Marketplace listing	High	This is the strongest price/model match of the group because the public sources explicitly name RMD-150 .

		brush/sand size.				
RSM Machinery	RMD-165	RSM's product page identifies the RMD-165 Series as a semi-automatic/manual-style middle-ground option for deburring parts from laser, plasma, or punch operations. Additional public listings also explicitly name RMD-165 .	No fixed public price verified for the exact RMD-165 .	Quote-only / no posted price found	Low	I could verify the model exists and find public spec-style pages, but not a trustworthy posted price tied specifically to RMD-165 .
Jinan D. Ventus	Manual flat-sheet deburring machine	D. Ventus has a public Made-in-China product page for a "Precision Manual Grinding	US\$8,000–18,500 / set FOB for D. Ventus industrial manual grinding/deburring machine	Marketplace listing	Medium	Useful public budget range, but I could not confirm a precise single model code

		Deburring Machine for Flat Sheets with Polishing Function.” The company storefront also shows broader sheet deburring/polishing machine offerings.	listings in its SS deburring and edge-rounding group.			from the snippet for the priced listing.
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Best current public price snapshot

If you just want the current **publicly visible numbers**:

- **Knoppo manual deburring machine: US\$3,999** public listing.
- **RSM RMD-150**: about **US\$3,500 FOB** public listing.
- **RSM RMD-165**: **no exact posted public price** verified.
- **D. Ventus manual deburring machine: US\$8,000–18,500 FOB** public range for related industrial manual deburring listings.

My take on which listings are most trustworthy

The **RSM RMD-150** price is the cleanest and most procurement-friendly public match because the sources explicitly tie the price/spec listing to **RMD-150**.

The **Knoppo** number is probably useful for budgeting, but I would not treat **US\$3,999** as a confirmed **KD-BB1308D** quote without asking the supplier directly, because the price snippet is tied to a generic Knoppo manual deburring listing rather than the KD-BB1308D product page itself.

The **D. Ventus** range is broad, which usually means configuration, width, and tooling affect price a lot.

Recommended budgeting bands

Based on the public listings I verified, a sensible first-pass budget is:

- **Entry manual Chinese machines: ~US\$3.5k–4k** for Knoppo/RSM-level public listings.
- **Larger industrial manual/flat-sheet deburring machines: ~US\$8k–18.5k** for D. Ventus-style listings.

If you want, I can do the next step and make a **supplier contact sheet** with model, public price, contact page, and the exact questions to send each vendor so you can get apples-to-apples quotes.

yes, please

I'm pulling the official contact pages and turning this into a quote-request sheet you can actually use, not just a vendor list.

Thought for 16s

Here's a supplier contact sheet you can actually use.

Supplier contact sheet

Supplier	Models to ask about	Official contact info	Public price signal	What to ask first
Jinan Knoppo Automation	KD-BB1308D Manual Deburring Machine	Email: info@knoppoauto.com ; WhatsApp/Tel: +86 18663717059 ; address shown on product pages in Jinan, Shandong, China.	Public marketplace signal around US\$3,999 for a Knoppo manual grinding/deburring machine listing, but I could not verify that this exact price is for KD-BB1308D specifically.	Confirm whether the quoted machine is exactly KD-BB1308D , what consumables are included, and whether dust extraction is included or separate.
RSM Machinery	RMD-150, RMD-165	Email: market2@rsm-machinery.com ; WhatsApp: +86	RMD-150 has the cleanest public price signal: around	Ask them to quote RMD-150 vs RMD-165 side by side , because

		13917564315 on the RMD-150 page; another RSM catalog/contact page also lists +86-18962685190 and the same email.	US\$3,500 FOB from marketplace listings tied to that model family. I could not verify a fixed public price for RMD-165.	RMD-150 is positioned as entry-level manual deburring and RMD-165 as the more premium / semi-automatic middle-ground option.
Jinan D. Ventus	Manual flat-sheet deburring machine; also ask for their closest model to your part size	Email: info@dventuscn.com; Mobile/WhatsApp: +86 13256664133; Jinan, Shandong address listed on the company site.	Public listings show a broad range, roughly US\$8,000–18,500 FOB for related deburring/edge-rounding machines, but the exact configuration matters a lot.	Ask them for the exact model code matching your use case, because their public listings span multiple widths and configurations.

Exact questions to send every supplier

Copy this list into your inquiry so you get apples-to-apples quotes:

1. "Please quote the exact model and include the **model code**."
2. "What is the **FOB price**, and what is the **CIF/DDP price to my destination**?"
3. "What is included in the quote: **machine, consumables, dust collector, spare parts, tooling, shipping crate, installation, training**?"
4. "Please confirm the machine's **working area / table size / belt or brush size / max part thickness**."
5. "Which materials is this configuration best for: **carbon steel, stainless steel, aluminum**?"
6. "Can it handle **laser-cut burrs, edge rounding, oxide removal, and light slag removal**?"
7. "What is the **power requirement and voltage**?"

8. “What **consumables** are required, how often do they wear out, and what are the replacement prices?”
9. “What is the **lead time** for production and shipment?”
10. “What is the **warranty period** and what parts are covered?”
11. “Do you provide **English manuals, video setup support, and remote troubleshooting?**”
12. “Please share **customer references, machine videos, and photos/videos processing parts similar to mine.**”

Those questions matter because the public pages show meaningful differences in positioning: RSM presents **RMD-150** as entry-level for small flat sheet parts and **RMD-165** as a higher-end manual/semi-automatic option, while D. Ventus appears to sell several broader deburring configurations rather than one simple manual model.

Best way to compare quotes

Tell every supplier the same part details:

- material: carbon steel / stainless / aluminum
- thickness range
- max part size
- typical burr type: laser burr, oxide, slag, sharp edge
- target finish: deburr only, edge rounding, polishing
- monthly volume
- power available in your shop

That matters because Knoppo’s KD-BB1308D page, RSM’s RMD pages, and D. Ventus’s listings all indicate different processing scopes and machine sizes.

My recommendation

Start with **RSM first** because it has the clearest public model split and the cleanest price trail for **RMD-150**. Then ask **Knoppo** for a direct KD-BB1308D quote, and ask **D. Ventus** to identify the closest exact model before comparing price.

I can also draft the actual supplier inquiry message for you.

okay, please

Thought for a few seconds

Here’s a clean supplier inquiry email you can send to **RSM, Knoppo, and D. Ventus**.

Email

Subject

Inquiry for Manual Deburring Machine QuoteInquiry for Manual Deburring Machine Quote

Hello,

I am looking for a manual deburring machine for metal parts and would like to request a quotation.

My application details are below:

- Material: [carbon steel / stainless steel / aluminum]
- Thickness range: [example: 1–10 mm]
- Maximum part size: [example: 1300 x 800 mm]
- Typical parts: [laser-cut flat parts / punched parts / welded parts]
- Main requirement: [deburring / edge rounding / oxide removal / light slag removal / polishing]
- Production volume: [example: low volume / medium volume / parts per month]
- Power available in my shop: [example: 380V 3-phase]

Please provide the following information:

1. Exact model name and model code
2. FOB price
3. CIF or DDP price to [your destination country/city]
4. Lead time
5. Machine specifications:
 - working area / table size
 - maximum workpiece thickness
 - abrasive type / brush / belt size
 - motor power
 - voltage requirement
6. What is included in the quote:
 - machine
 - consumables
 - dust collector
 - spare parts

- installation tools
- training

7. Recommended configuration for my application
8. Consumable lifetime and replacement cost
9. Warranty period and after-sales support
10. English manual availability
11. Videos or photos of the machine processing parts similar to mine

I am currently comparing the following models or equivalent:

- RSM RMD-150 / RMD-165
- Knoppo KD-BB1308D
- D. Ventus manual deburring machine

Please also let me know which model you recommend for my use case and why.

Thank you. I look forward to your reply.

Best regards,

[Your name]

[Company name, if applicable]

[Country]

[WhatsApp / phone]

Here's a shorter version for WhatsApp or Alibaba chat:

Message

Hello, I am looking for a manual deburring machine for metal parts.

My application:

- Material: []
- Thickness: []
- Max part size: []
- Process needed: deburring / edge rounding / oxide removal / polishing
- Volume: []
- Power: []

Please send:

1. Exact model and specs

2. FOB price
3. Shipping price to [destination]
4. Lead time
5. What is included in the machine
6. Consumable cost
7. Warranty
8. English manual
9. Video of similar parts being processed

Please recommend the best model for my application.

Thank you.

Fill in your part details and I'll tailor it for your exact use case.